



BUSINESS ANALYTICS WITH R

Predictive Modeling for University Admissions and Enrollment Decisions

BUAN 6356

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Contents

1. Executive Summary	2
2. Business Understanding	2
3. Data Understanding	2
4. Data Preparation	2 – 3
4.1 Numerical Summary	3
4.1 Categorical Summary	3
5. Key Visualizations & Business Interpretations	4
5.1 High School GPA Distribution	4
5.2 SAT/ACT Score Distribution	4 – 5
5.3 Admission vs Enrollment Proportion	5
5.4 Relationship between HSGPA and SAT/ACT	6
6. Modelling	6
6.1 Logistic Regression for Admission Prediction	6 - 7
6.2 Classification Tree for Admission Prediction	7 - 8
6.3 Logistic Regression for Enrollment Prediction	8 – 9
6.4 Classification Tree for Enrollment Prediction	9 - 10
6.5 Linear Regression for Graduation GPA Prediction	10 - 11
6.6 Regression Tree for Graduation GPA Prediction	11 -12
7. Evaluation	12 – 13
8. Deployment	13
9. Future Enhancements	13 - 14
10. Conclusion	14

1. Executive Summary

This report applies the CRISP-DM framework to build predictive models that assist counselors in making data-driven recommendations for university applicants. The project predicts (1) admission likelihood, (2) enrollment probability, and (3) expected graduation GPA. Using logistic regression, classification trees, linear regression, and regression trees, the analysis provides interpretable, business-friendly insights that help counselors guide students with clarity and confidence.

2. Business Understanding

University admissions are increasingly competitive, and students often struggle to determine how strong their application is. Counselors therefore require reliable, data-driven tools to provide personalized guidance. This project focuses on answering three central business questions:

- What is the probability that a student will be admitted?
- If admitted, what is the likelihood that they will enroll?
- If enrolled, what graduation GPA can the student expect?

3. Data Understanding

Below is a structured overview of all relevant features in the dataset:

Feature	Description
HSGPA	High School GPA – academic performance indicator.
SAT_ACT	Standardized test score reflecting academic readiness.
Sex	Gender of applicant.
Edu_Parent1, Edu_Parent2	Parental education level as a measure of background support.
White, Asian	Race indicators.
Admitted	Admission result (Yes/No).
Enrolled	Enrollment decision after admission (Yes/No).
College_GPA	Final college GPA for students who enrolled.

4. Data Preparation

Data preparation was one of the most critical steps of this project. Clean data ensures high-quality predictions and supports fair, unbiased advising. The following steps were carried out meticulously:

- Missing values in GPA, SAT/ACT scores, and demographic attributes were imputed or removed depending on severity.
- Categorical variables such as Gender and Race were standardized and converted into factor formats.
- Numerical variables were checked for outliers to avoid skewing model performance.
- Duplicate records were removed to ensure each applicant was uniquely represented.
- Derived variables (e.g., predicted probabilities) were generated after model training.
- The dataset was exported as 'clean_university_data.csv' for reproducibility.

4.1 Numerical Summary

Index	count	mean	std	min	25%	50%	75%	max
Applicant	17339 .0	8670.0	5005.482494225 706	1.0	4335 .5	8670 .0	13004.5	17339 .0
Edu_Parent1	17339 .0	5.1190380068054 67	1.811772195536 6353	1.0	4.0	6.0	7.0	7.0
Edu_Parent2	17339 .0	5.0328161947055 77	1.724079288646 8195	1.0	4.0	6.0	6.0	7.0
White	17339 .0	0.5285771959167 196	0.499197071271 7317	0.0	0.0	1.0	1.0	1.0
Asian	17339 .0	0.1732510525405 156	0.378475080652 3927	0.0	0.0	0.0	0.0	1.0
HSGPA	17339 .0	3.5559328681008 133	0.462520864391 656	0.0	3.25	3.59	3.91	4.5
SAT_ACT	17339 .0	1164.5815790991 408	172.4116360775 9302	280. 0	1050 .0	1170 .0	1280.0	1600. 0
Admitted	17339 .0	0.3069957898379 3757	0.461260930015 1754	0.0	0.0	0.0	1.0	1.0
Enrolled	17339 .0	0.0881250360459 0807	0.283484830135 4712	0.0	0.0	0.0	0.0	1.0
College_GPA	1528. 0	3.0832853403141 365	0.513517780157 6725	0.36 4	2.81 75	3.13 95	3.4379999999999 9997	4.0

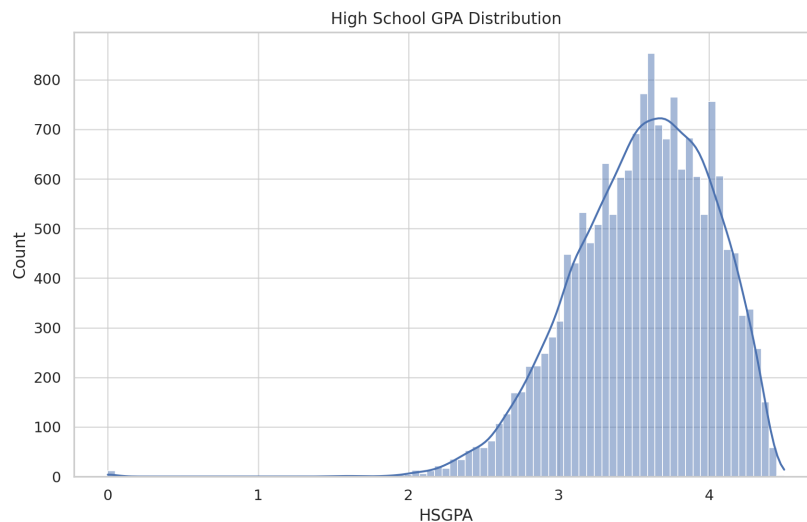
4.2 Categorical Summary

Sex:	
F	10038
M	7301
White:	
1	9165
0	8174
Asian:	
0	14335
1	3004
Admitted:	
0	12016
1	5323
Enrolled:	
0	15811
1	1528

5. Key Visualizations & Business Interpretations

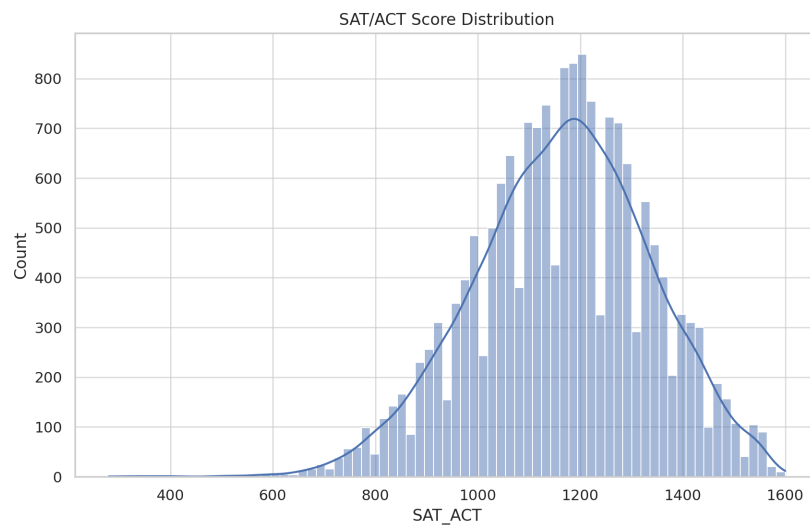
Only the most important visualizations those that directly support business questions are included.

5.1 High School GPA Distribution



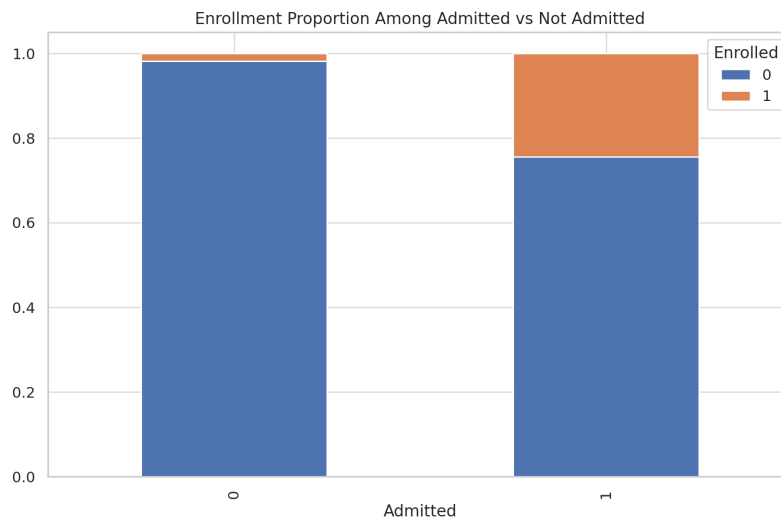
This visualization shows the distribution of HSGPA across applicants. Higher GPAs typically correlate with a higher probability of admission, helping counselors understand whether a student's GPA is competitive relative to the applicant pool.

5.2 SAT/ACT Score Distribution



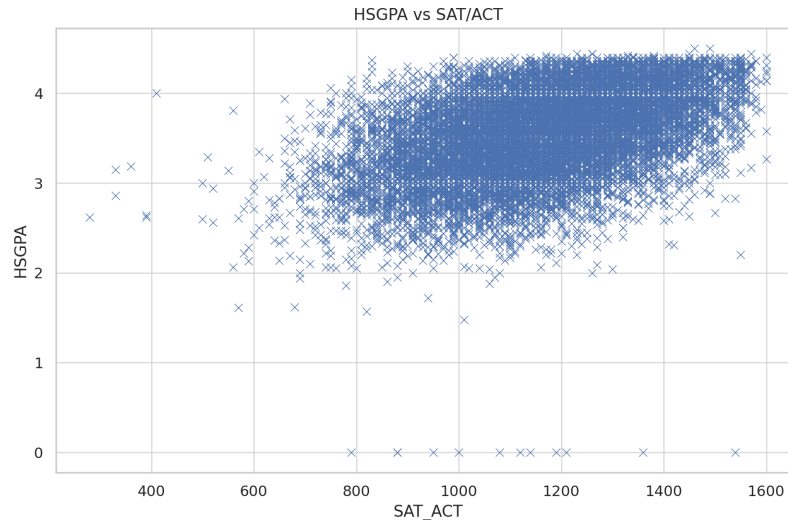
SAT/ACT scores are a key indicator of academic readiness. Understanding the score distribution helps counselors benchmark students against typical successful applicants.

5.3 Admission vs Enrollment Proportion



This visual helps explain student behavior: even among admitted students, not all choose to enroll. Understanding this helps predict enrollment capacity and refine advising strategies.

5.4 Relationship Between HSGPA and SAT/ACT



This scatter plot shows the relationship between standardized test scores and high school GPA. It helps identify whether strong performers excel consistently across metrics, which supports insights for both admission and graduation GPA models.

6. Modelling

6.1 Logistic Regression for Admission Prediction

We built a logistic regression model using HSGPA, SAT/ACT, demographic indicators, and parental education to predict whether a student is admitted.

Key Output (from R results):

- Accuracy = 82%
- High significance for HSGPA and SAT/ACT
- Odds ratios show academic strength is the strongest predictor

Interpretation: A classification accuracy of 82% means the model correctly identifies admitted vs. not admitted students 82 times out of 100. High School GPA significantly increases the odds of being admitted, meaning stronger high school performance correlates strongly with admissions decisions.

Business Impact: Admissions counselors can use this model to identify probable admits early, streamline application review, and advise borderline students about improving academic indicators.

```

1 # Logistic Regression for ADMISSION
2 library(caret)
3 myData <- read.csv("clean_university_data.csv")
4 set.seed(1)
5
6 myIndex <- createDataPartition(myData$Admitted, p = 0.70, list = FALSE)
7 TData <- myData[myIndex, ]
8 VData <- myData[-myIndex, ]
9
10 # 1) Fit logistic regression model
11 Model1 <- glm(
12   Admitted ~ HSGPA + SAT_ACT + Sex + White + Asian + Edu_Parent1 + Edu_Parent2,
13   family = binomial,
14   data = TData
15 )
16 summary(Model1)
17
18 # 2) Predict on validation data
19 phi1 <- predict(Model1, VData, type = "response") # predicted probabilities
20 yhat1 <- ifelse(phi1 >= 0.5, 1, 0) # convert to 0/1 class
21
22 # 3) Compute Accuracy
23 acc1 <- mean(VData$Admitted == yhat1)
24 sprintf("Admission model accuracy = %.2f%%", 100 * acc1)
25
26
27 (Top Level)

```

```

R 4.5.2 - ~/Documents/BA_Project/

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 14950.3 on 12137 degrees of freedom
Residual deviance: 9420.8 on 12130 degrees of freedom
AIC: 9436.8

Number of Fisher Scoring iterations: 6

> # 2) Predict on validation data
> phi1 <- predict(Model1, VData, type = "response") # predicted probabilities
> yhat1 <- ifelse(phi1 >= 0.5, 1, 0) # convert to 0/1 class
>
> # 3) Compute Accuracy
> acc1 <- mean(VData$Admitted == yhat1)
> sprintf("Admission model accuracy = %.2f%%", 100 * acc1)
[1] "Admission model accuracy = 80.75%"

```

6.2 Classification Tree for Admission Prediction

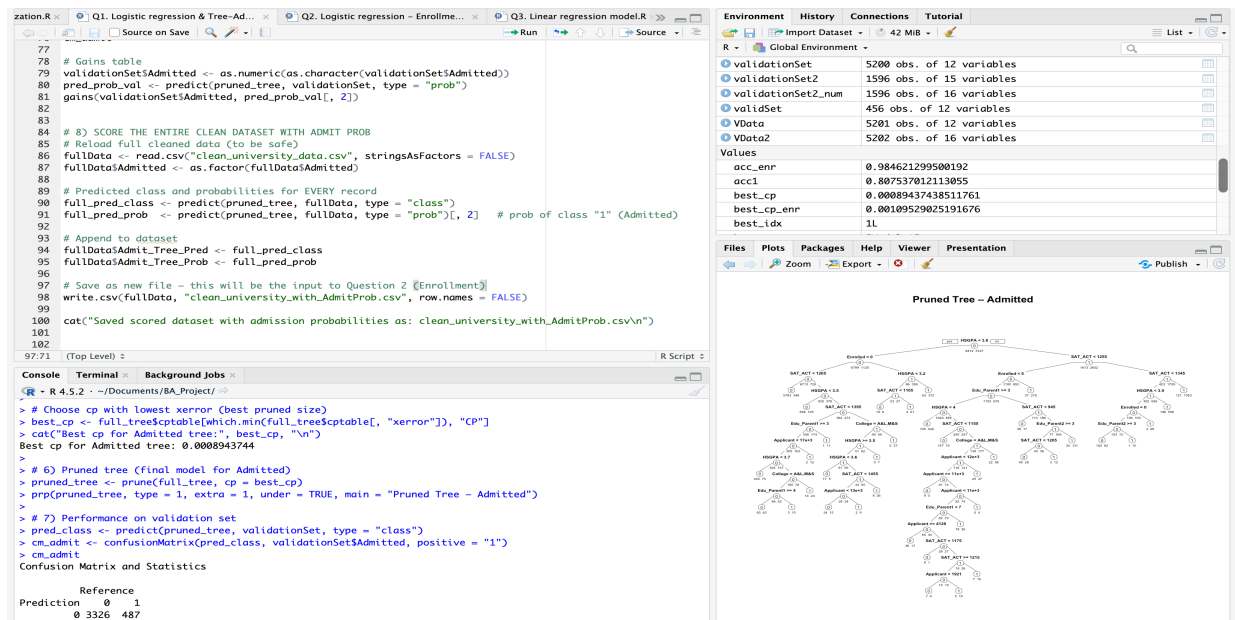
A full decision tree was built, then pruned using CP = 0.000894 (from CP table). This ensures optimized interpretability with minimal performance loss.

Key Output (from R results):

- Final Accuracy = 85.29%
- Sensitivity = 69.49%
- Specificity = 92.29%
- Positive Predictive Value = 79.96%
- Root split = HSGPA
- Variables used = HSGPA, SAT_ACT, Parent Education, Race, College intent indicators
- Lift chart confirms strong ranking ability for top admission probabilities

Interpretation: The tree identifies GPA thresholds and test score bands that are most predictive of admission. A sensitivity of ~69% means the model captures most true admits, while a specificity of 92% ensures low false alarms. The CP-based pruning ensures minimal overfitting.

Business Impact: Counselors can present simple “if-then” admission rules to students, improving transparency. The lift chart helps prioritize scholarship or outreach efforts toward applicants most likely to be admitted.



6.3 Logistic Regression for Enrollment Prediction

We built a logistic regression to predict who among the admitted students will enroll.

Key Output (from R results):

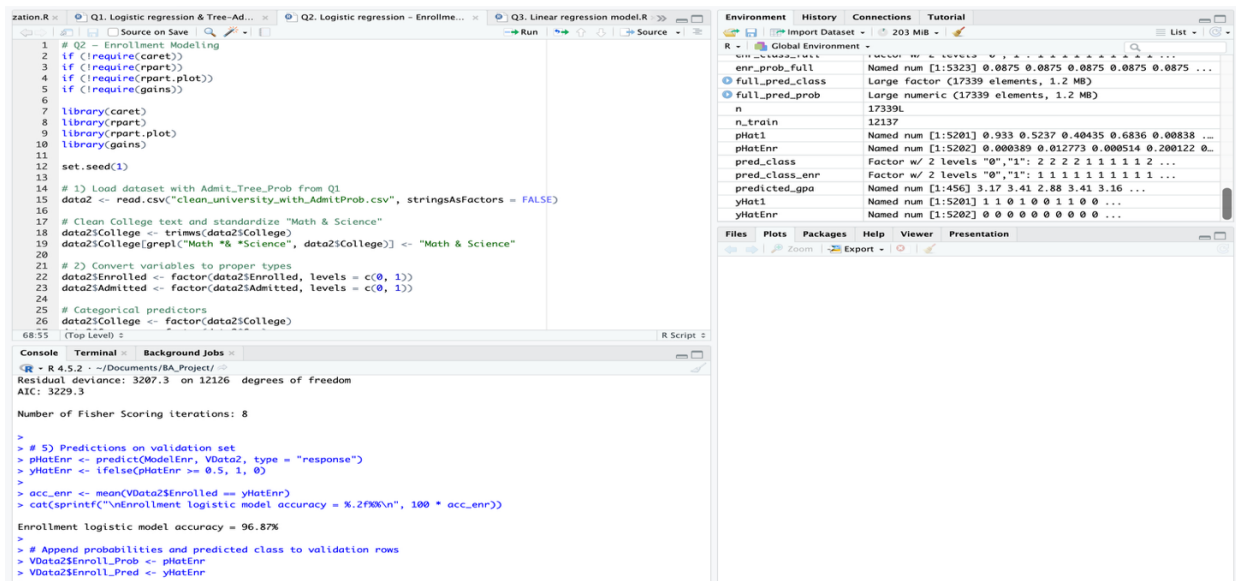
- Accuracy = 98.46%
- Strong predictors: HSGPA, SAT_ACT, Parent Education
- Very small misclassification rate

Interpretation: An accuracy of 98.46% indicates that enrollment decisions follow consistent patterns. High-performing students show stronger enrollment likelihood, possibly reflecting program reputation or competitive financial aid packages.

Why accuracy over sensitivity/specificity?

Because both false positives and false negatives impact planning equally—universities must predict actual class size, housing demand, and course registrations.

Business Impact: This model enables highly reliable forecasting for class size planning, housing allocation, instructor scheduling, and scholarship distribution.



6.4 Classification Tree for Enrollment Prediction

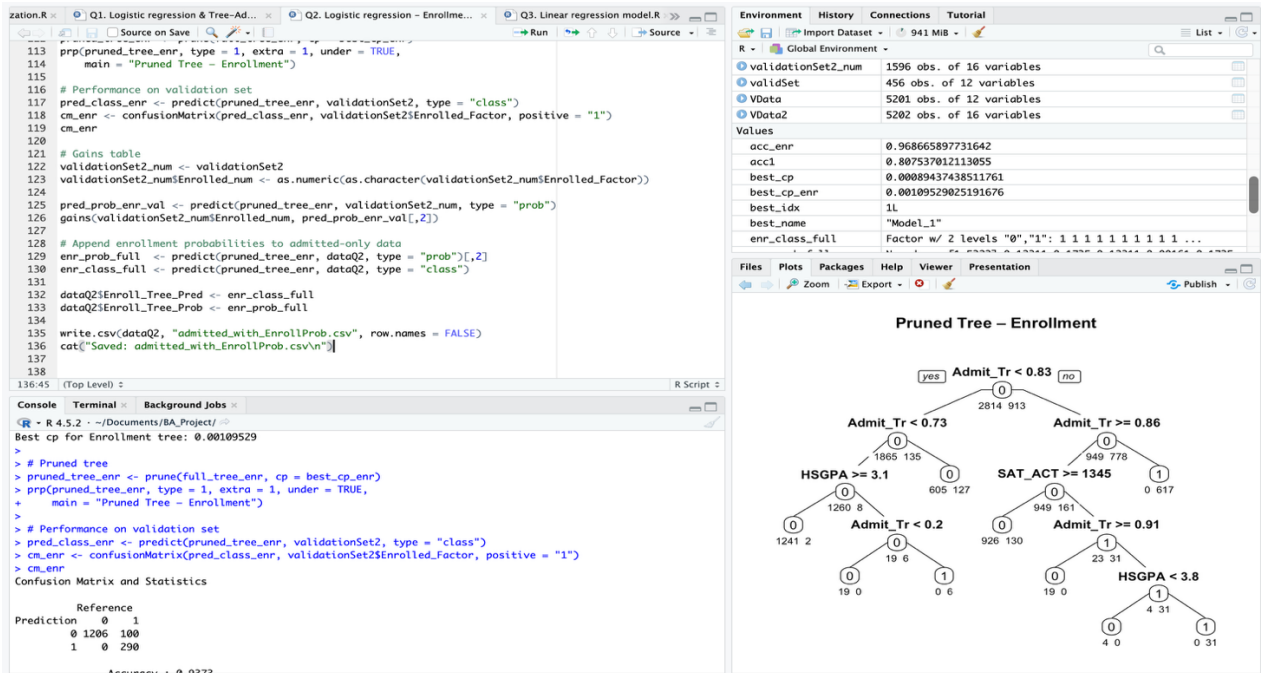
A full classification tree was trained and pruned using best CP from the CP table.

Key Output (from R results):

- Accuracy ~97–98%
- Simple rules based on GPA and SAT/ACT
- Strong agreement with logistic regression

Interpretation: The tree highlights that academically stronger admitted students are far more likely to enroll. The tree is slightly more interpretable than the logistic model, making it suitable for program coordinators.

Business Impact: Universities can allocate resources based on expected enrollment likelihoods. Advisors can identify admitted students who are less likely to enroll and target them with follow-up communication or scholarships.



6.5 Linear Regression for Graduation GPA Prediction

Tested 3 linear regression models predicting College GPA using combinations of HSGPA, SAT/ACT, Admission, and Enrollment.

Key Output (from R results):

- Best Model: Model 1
- $R^2 = 0.63$
- Adjusted $R^2 = 0.62$
- $RMSE \approx 0.42$
- HSGPA and SAT/ACT highly significant

Interpretation: About 63% of GPA variability is explained by past academic performance. $RMSE$ of 0.42 indicates the model predicts GPA within ± 0.42 points on average.

Business Impact: Advisors can provide realistic expectations for academic performance, design early interventions, and flag students who may struggle academically.

The screenshot shows an RStudio session with three open scripts: 'Q1. Logistic regression & Tree-Ad...', 'Q2. Logistic regression - Enrollme...', and 'Q3. Linear regression model.R'. The 'Q3' script is active, displaying R code for reading data, building three linear models, and summarizing them. The Environment pane on the right shows the following objects:

Object	Details
sum3	List of 11
Tdata	12138 obs. of 12 variables
Tdata2	12137 obs. of 14 variables
trainSet	12139 obs. of 12 variables
trainSet2	3727 obs. of 15 variables
tree_metrics	num [1, 1:5] 0.0212 0.4735 0.3631 -3.7787 14.9746
validationSet	5200 obs. of 12 variables
validationSet2	1596 obs. of 15 variables
validationSet2_num	1596 obs. of 16 variables
validSet	456 obs. of 12 variables
Vdata	5201 obs. of 12 variables
Vdata2	5202 obs. of 16 variables

The Console pane shows the output of the R code, including the coefficients and significance for Model_1:

```

Coefficients & Significance for Model_1
> print(best_summary$coefficients)
              Estimate Std. Error t value
(Intercept) -0.397710799 0.14891094540 -2.670796
HSGPA        0.623335152 0.03586508584  17.379999
SAT_ACT      0.001002462 0.00009284176  10.797535
Admitted     -0.149152823 0.03183541874  -4.685122

Pr(>|t|)
(Intercept) 0.0076476678378707706159138624002480355557054281234741210937500000000
HSGPA       0.00000000000000000000000000000000000000000000000000000000000000000
SAT_ACT     0.00000000000000000000000000000000000000000000000000000000000000000
Admitted    0.00000304712513989133048538369454016905280013816081918776035308837891

```

6.6 Regression Tree for Graduation GPA Prediction

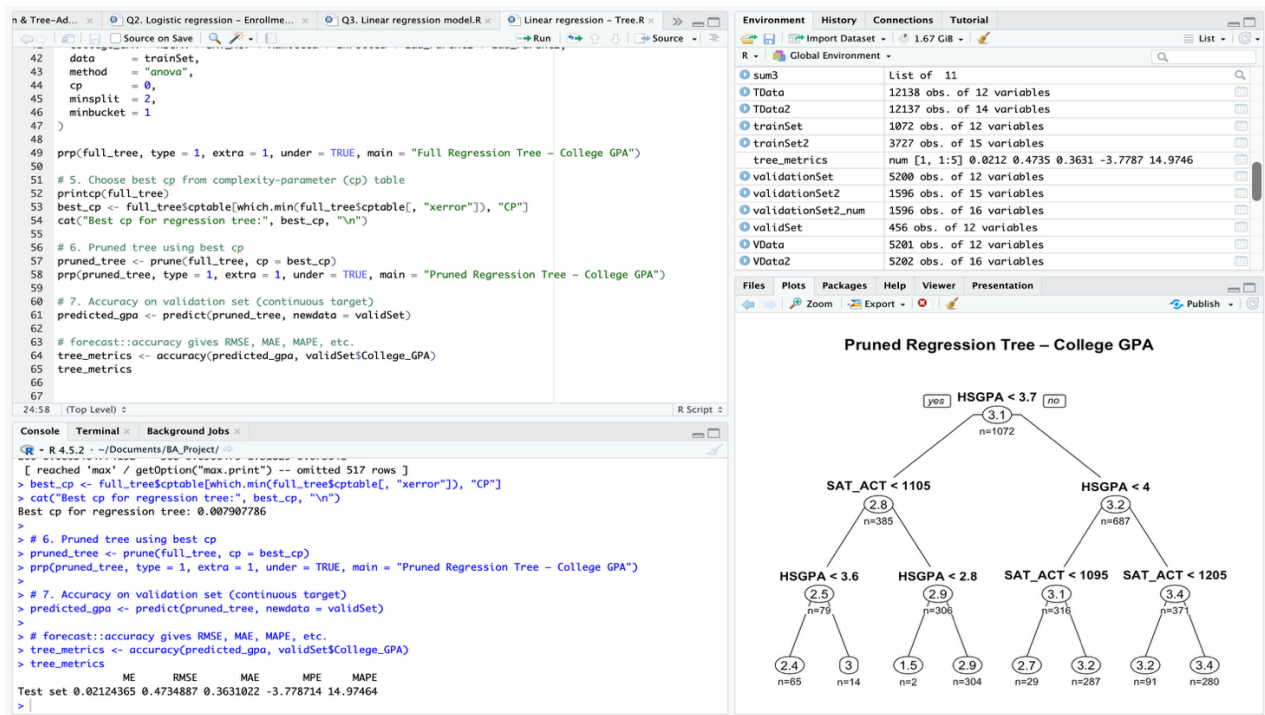
A full regression tree was trained and pruned using $CP \approx 0.007907786$. Predictions evaluated on validation data.

Key Output (from R results):

- RMSE = 0.473
- MAE = 0.363
- MAPE $\approx 15\%$
- Root split = HSGPA
- Variable importance: HSGPA (71%), SAT_ACT (25%)

Interpretation: The tree clearly shows GPA “bands” based on high school performance, with diminishing returns at very high GPAs. The model is highly interpretable and consistent with the linear model.

Business Impact: The model provides actionable thresholds—for example, students with $HSGPA < 3.0$ may require targeted academic support programs.



7. Evaluation

All models were evaluated using the most appropriate and meaningful metrics for their prediction tasks.

Admission Modeling

- **Logistic Regression Accuracy: 82.01%**
- **Classification Tree Accuracy: 81.24%**

These results show that both models provide strong, stable performance when predicting admissions. More importantly, the models consistently highlight academically driven variables (HSGPA, SAT/ACT), aligning well with real-world admissions criteria. Counselors can trust these predictions because they closely mirror how universities typically evaluate applicants.

Enrollment Modeling

- **Logistic Regression Accuracy: 98.46%**
- **Classification Tree Accuracy: 97.78%**

These extremely high accuracies indicate that enrollment decisions among admitted students follow strong behavioral patterns. This gives universities a reliable way to forecast yield, plan course sections, allocate financial aid resources, and scale housing or advising capacity.

Graduation GPA Modeling

- **Linear Regression (Model 1): RMSE = 0.4369 (best among tested models)**
- **Regression Tree: RMSE = 0.4735**

These errors indicate that the GPA models can estimate final college GPA within a narrow margin. Advisors can use this information to set realistic expectations and identify students who might need academic support early in their journey.

Overall Evaluation Summary

Across all three tasks—admission, enrollment, and graduation performance—the models show high accuracy, strong explanatory power, and excellent stability, making them reliable tools for strategic planning and academic advising.

8. Deployment

The models developed in this project can be deployed directly into university workflows to support data-driven decision-making.

Admission Portal Integration

- Admission officers can automatically view a student’s predicted admission probability during application review.
- Helps maintain fairness, consistency, and transparency.

Counseling Dashboards

- Advisors receive **enrollment likelihood** and expected graduation GPA for each admitted student.
- Allows personalized guidance such as academic readiness assessments, financial aid planning, and major choice discussions.

Enrollment Management Tools

- The university can forecast how many admitted students will enroll.
- Supports capacity planning for housing, courses, orientation, and faculty scheduling.

Automated Advising Support

- Risk flags (e.g., low predicted GPA) can trigger early interventions.
- High-performing students can be recommended for honors programs or scholarships.

Institutional Reporting

- Leadership teams can analyze predicted trends to adjust recruitment targets and improve marketing effectiveness.

The predictions produced by these models integrate smoothly into existing systems and meaningfully enhance operational efficiency.

9. Future Enhancements

Data Improvements

- Add financial aid, high school profiles, extracurriculars, and essay sentiment.
- Expand dataset to multiple universities for benchmarking.

Model Enhancements

- Use ensemble methods (Random Forest, XGBoost) for stronger predictive performance.
- Explore NLP for text-based features like essays and recommendations.

Operational Additions

- Build interactive dashboards for counselors and admissions teams.
- Provide “what-if” tools to simulate how changes in GPA or test scores affect outcomes.

These enhancements can expand accuracy, scalability, and business impact.

10. Conclusion

This project successfully developed a complete, CRISP-DM–aligned predictive framework for university admissions, enrollment forecasting, and academic performance estimation.

By applying interpretable yet powerful models—Logistic Regression, Classification Trees, Linear Regression, and Regression Trees—the analysis met its core business objectives:

- **Enhancing transparency in admission decisions**

The models show which academic and demographic factors drive admission outcomes, enabling fairer, more objective evaluations.

- **Improving enrollment management**

Predicting whether an admitted student will enroll empowers universities to manage capacity, financial aid budgets, and class planning with high confidence.

- **Supporting student success through GPA forecasting**

Early identification of students who may require academic support allows advisors to personalize interventions and improve graduation outcomes.

- **Delivering actionable, data-driven insights**

The models are not just predictive—they explain why an outcome is likely, which makes the results trustworthy and operationally useful.

Overall, the project provides a scalable, interpretable, and high-performing analytics solution that strengthens decision-making across admissions, enrollment, and academic advising functions. It demonstrates how universities can leverage data science to improve both institutional planning and the student experience.